

Bidirectional Logarithmic Spiral Interference as a Geometric Substrate for Emergent Gravity

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Abstract—This paper proposes a geometric field model in which gravitational-like organization may emerge from a bidirectional logarithmic spiral interference lattice. Two oppositely oriented logarithmic spirals sharing a common origin are treated as phase-carrying trajectories within a dynamical field. Their superposition produces a recursive network of intersections, constructive interference nodes, and curvilinear transport pathways. These features are interpreted as coherent structural regions within an underlying pre-geometric medium. While the model remains speculative and does not derive known gravitational field equations, it provides a structured framework for exploring how spatial organization, stability, and effective attraction might arise from interference geometry. Potential simulation approaches and testable implications are discussed.

Index Terms—Emergent gravity, logarithmic spiral, interference lattice, geometric physics, field coherence, self-similarity, dynamical systems

I. INTRODUCTION

Gravity is conventionally described as the curvature of spacetime in general relativity [1], while modern theory also treats it as an effective field theory below a cutoff scale [8]. Several approaches suggest gravity may be emergent rather than fundamental. Thermodynamic arguments relate gravitational dynamics to entropy and local equilibrium [2]–[5]. Complementary models propose entropic or induced mechanisms [6], [7], and analogue-gravity systems demonstrate that metric-like behavior can arise in non-gravitational media [9], [10].

More recent work connects spacetime structure to quantum information and entanglement [11], [12]. These perspectives motivate exploration of geometric or wave-based substrates capable of producing large-scale organization.

This paper introduces a candidate substrate based on interference between two counter-oriented logarithmic spirals. The goal is not to replace general relativity, but to explore whether structured interference geometry can produce persistent organization and effective attraction.

II. MATHEMATICAL FOUNDATION

A. Logarithmic Spiral Geometry

A logarithmic spiral is defined by

$$r(\theta) = ae^{b\theta}, \quad (1)$$

where $a > 0$ sets scale and b controls growth. The spiral exhibits self-similarity and continuous curvature, making it suitable for multiscale organization [13], [14].

B. Bidirectional Spiral Construction

Two oppositely oriented spirals are defined as:

$$r_1(\theta) = ae^{b\theta}, \quad (2)$$

$$r_2(\theta) = ae^{-b\theta}. \quad (3)$$

C. Field Representation

The field Ψ may be interpreted as an effective wavefunction-like quantity encoding amplitude and phase of underlying dynamical modes. We model the system using a complex scalar field $\Psi(r, \theta, t)$, and the total field is

$$\Psi(r, \theta, t) = \Psi_1(r, \theta, t) + \Psi_2(r, \theta, t). \quad (4)$$

A minimal dynamical model may be introduced using a wave-like equation:

$$\frac{\partial^2 \Psi}{\partial t^2} = c^2 \nabla^2 \Psi + V(r, \theta) \Psi, \quad (5)$$

where $V(r, \theta)$ represents an effective potential encoding geometric constraints imposed by the spiral trajectories, guiding phase evolution along logarithmic paths. This equation serves only as a candidate structure for future refinement.

As illustrated in Fig. 1, the superposition of counter-oriented spirals produces a structured interference lattice with localized high-coherence nodes.

III. PHYSICAL INTERPRETATION

Constructive interference regions correspond to coherent nodes. Such interference-based structuring is consistent with known wave behavior, where superposition of coherent waves produces stable patterns of constructive and destructive interference. Similar structures arise in wave interference and singularity formation [15], [16], where standing-wave patterns generate stable nodal configurations. These nodes may act as persistent structural features within the field. Here, coherence density is quantified locally as the squared magnitude of the superposed field, i.e., $|\Psi(r, \theta)|^2$, serving as a direct measure of constructive interference strength.

Gradients in node density and field coherence define preferred pathways through the system. Test particles or effective excitations propagating through the field are assumed to follow effective trajectories determined by these gradients, producing a transport bias toward regions of higher coherence.

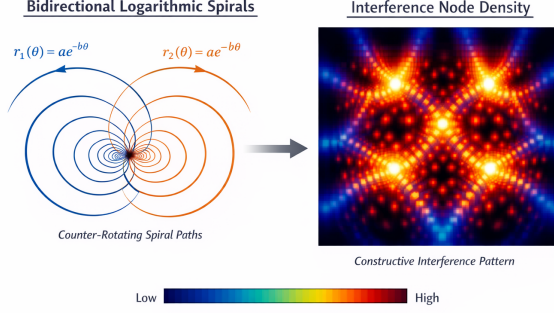


Fig. 1. Bidirectional logarithmic spiral construction and resulting interference node density. Left: two counter-oriented logarithmic spirals defined by $r_1(\theta) = ae^{b\theta}$ and $r_2(\theta) = ae^{-b\theta}$. Right: example interference pattern illustrating constructive node formation and coherence gradients. Color scale represents relative field intensity $|\Psi|^2$.

IV. EMERGENT GRAVITY INTERPRETATION

Under this framework, gravitational-like behavior (specifically, effective attractive behavior) may arise from:

- gradients in coherence density,
- recursive reinforcement of interference nodes,
- path-dependent transport through the lattice.

This interpretation aligns conceptually with emergent gravity approaches in which macroscopic dynamics arise from deeper statistical or geometric structure [4]–[6].

V. SIMULATION APPROACH

A computational program includes:

- 1) generating spiral fields,
- 2) assigning phase evolution,
- 3) computing interference patterns,
- 4) identifying nodes and coherence gradients,
- 5) simulating effective particle trajectories.

Nodes are detected computationally as local maxima of $|\Psi(r, \theta)|^2$ exceeding a chosen threshold, optionally combined with phase-vortex criteria. Standard numerical methods such as finite-difference, spectral, and FFT-based techniques may be applied [17]–[20].

These simulations provide a means to evaluate whether stable node clustering and trajectory curvature emerge robustly across parameter regimes. A key question is whether stable node clustering emerges dynamically rather than being imposed by initial conditions.

VI. LIMITATIONS

This model does not derive Einstein’s field equations, establish relativistic invariance, or define a complete physical interpretation of Ψ . It remains a speculative geometric hypothesis. In particular, the proposed dynamics are illustrative rather than derived from a fundamental action principle. The model does not currently specify boundary conditions or conservation

laws for Ψ , which would be required for a complete physical theory.

VII. CONCLUSION

Bidirectional spiral interference provides a structured candidate framework for emergent gravitational organization. By combining scale-invariant geometry with interference-based coherence, the model offers a testable pathway for exploring how effective attraction may arise from underlying field structure.

Establishing a rigorous connection to known physical theories remains an open challenge and a key direction for future work, particularly through numerical simulation, formal dynamical development, and comparison with analogue physical systems.

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